

MARION COUNTY AP, AR, USA

WMO: 723447

Lat: 36.291N

Lon: 92.590W

Elev: 719

StdP: 14.32

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 03918

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.4	18.1	0.2	5.7	18.4	3.5	6.7	21.6	21.9	48.1	19.5	39.7	4.2	310	0.394

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	96.8	75.8	93.1	75.0	91.0	74.6	78.7	90.7	77.6	89.7	76.6	87.9	6.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.2	136.0	85.2	74.0	130.4	83.5	73.0	125.8	82.3	42.8	90.5	41.5	90.2	40.4	87.5	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.0	15.6	12.6	DB	5.7	100.2	4.9	2.6	2.3	102.0	-0.6	103.5	-3.3	105.0	-6.8	106.9	(4)
(5)			WB	5.1	80.2	4.1	1.5	2.1	81.3	-0.3	82.2	-2.6	83.1	-5.6	84.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.5	37.6	41.2	50.3	59.5	68.6	76.7	79.0	79.2	72.2	59.5	48.8	40.4	(6)	
	DBStd	17.11	10.90	11.58	9.77	8.92	7.06	4.74	4.61	5.03	6.20	8.81	9.25	9.38	(7)	
	HDD50	1308	405	294	117	23	1	0	0	0	0	15	130	323	(8)	
	HDD65	3724	850	669	463	206	51	0	0	0	14	220	488	763	(9)	
	CDD50	4777	20	47	126	307	576	801	898	904	666	310	95	27	(10)	
	CDD65	1717	0	1	7	41	162	351	433	439	229	50	3	1	(11)	
	CDH74	17085	0	16	82	441	1359	3536	4521	4500	2080	510	40	0	(12)	
	CDH80	7371	0	1	11	95	410	1533	2131	2200	833	154	3	0	(13)	
	WSAvg	4.7	5.8	6.0	6.3	6.3	4.2	3.7	3.1	3.1	3.1	4.1	4.8	5.3	(14)	
	PrecAvg	47.4	3.2	3.1	4.3	4.8	4.9	4.0	3.5	2.8	4.7	3.7	4.7	3.4	(15)	
(16) Precipitation	PrecMax	66.8	5.9	6.1	15.3	9.3	10.9	7.6	7.7	5.8	9.8	10.9	12.4	5.6	(16)	
	PrecMin	36.8	0.6	0.7	0.9	1.7	1.5	1.6	1.7	0.3	0.9	0.6	1.2	0.8	(17)	
	PrecStd	8.0	1.6	1.5	3.1	2.0	2.0	1.6	1.7	1.3	2.7	2.8	3.0	1.4	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.4	76.6	81.4	88.0	90.8	97.1	99.3	100.4	93.4	88.4	79.3	70.5	(19)	
		MCWB	56.4	60.2	61.5	67.1	71.4	76.5	76.3	75.4	73.6	71.8	64.4	59.0	(20)	
	2%	DB	65.7	70.4	75.3	82.0	87.9	93.0	95.3	97.3	91.1	84.2	73.3	65.9	(21)	
		MCWB	56.5	58.0	60.2	63.3	71.7	75.0	76.4	75.3	73.3	70.1	59.6	57.4	(22)	
	5%	DB	61.2	65.6	72.2	79.3	84.2	90.6	92.7	93.4	89.3	80.6	70.1	61.2	(23)	
		MCWB	51.2	55.3	57.9	62.1	70.9	74.6	75.9	74.6	72.9	66.5	59.0	52.0	(24)	
	10%	DB	55.0	61.3	68.0	75.0	81.6	88.4	90.5	90.9	85.0	74.9	65.6	55.4	(25)	
		MCWB	46.5	52.0	55.9	60.7	68.6	73.8	75.3	74.7	71.7	64.0	56.5	48.8	(26)	
	0.4%	WB	61.4	63.8	65.8	69.3	75.9	79.1	80.2	80.4	77.5	74.7	67.6	62.6	(27)	
		MCDB	67.1	73.5	77.0	81.0	86.7	92.0	91.4	92.6	89.1	84.7	76.2	68.1	(28)	
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	58.7	60.4	62.5	67.0	74.0	77.6	78.7	78.6	75.9	72.1	63.6	58.8	(29)	
		MCDB	64.2	68.2	73.4	77.4	84.1	89.9	90.7	90.5	86.5	81.7	70.5	65.2	(30)	
	5%	WB	52.5	57.3	60.3	65.0	72.4	76.4	77.8	77.3	74.6	69.3	60.2	53.8	(31)	
		MCDB	59.0	64.2	68.7	75.2	81.6	87.7	89.6	89.9	84.4	76.1	67.4	58.4	(32)	
	10%	WB	47.1	52.8	57.8	63.0	70.6	75.1	76.7	76.0	73.1	66.3	57.0	50.3	(33)	
		MCDB	54.2	59.9	65.7	72.8	79.0	85.4	87.7	87.3	82.0	73.2	64.0	55.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	19.6	20.5	22.0	22.6	20.6	20.4	20.4	20.8	21.3	22.9	22.9	19.0	(35)	
		MCDBR	28.0	28.2	27.5	27.1	24.3	23.5	22.7	25.0	25.5	27.1	29.0	26.0	(36)	
		MCWBR	18.6	18.6	15.9	13.5	11.3	9.4	8.4	8.5	10.3	13.7	18.2	18.3	(37)	
		MCDBR	23.1	23.6	23.4	21.6	20.8	20.9	20.3	21.4	21.0	22.9	23.1	22.3	(38)	
	5% WB	MCWBR	17.9	17.8	15.6	12.8	10.7	9.2	8.5	8.3	9.5	12.9	17.0	18.0	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.309	0.315	0.350	0.386	0.421	0.438	0.468	0.466	0.398	0.353	0.323	0.301	(40)	
	taud		2.494	2.474	2.413	2.316	2.277	2.244	2.217	2.195	2.371	2.479	2.514	2.533	(41)	
	Ebn at Noon		282	294	291	285	274	269	259	258	272	276	274	278	(42)	
	Edh at Noon		26	30	35	40	43	44	45	45	36	30	25	23	(43)	
(44) All-Sky Solar Radiation	RadAvg		770	950	1289	1630	1790	2044	1968	1803	1542	1164	832	658	(44)	
	RadStd		65	118	83	104	156	142	116	119	122	142	92	83	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page